International and Monetary Economics

Course Code: 30054

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Lecture 11

Roadmap

1. From fixed to variable output
   - Net exports: The Marshall-Lerner condition & J-Curve
2. Aggregate demand
3. Goods market equilibrium
4. The DD schedule
5. The AA schedule
6. Policy experiments
7. Liquidity trap

Net Exports

When does NX depend on q?

- Consider trade between Italy (home country) and the UK (foreign country)
- Italian net exports are

$$NX = Ex - Im$$

  – *NX, Ex, Im* are **measured in units of Italian (domestic) output**

- Trade flows and prices

$$Ex = Ex\left(\frac{P^*}{P^{\text{Im from ITA in £}}}\right) = Ex\left(\frac{P^*}{P/E}\right) = Ex\left(\frac{P^*E}{P}\right) = Ex(q)$$

$$Im = \frac{P^{\text{Im from UK in €}}}{P} Ex^*\left(\frac{P}{P^{\text{Im from UK in €}}}\right) = \frac{P^*E}{P} Ex^*\left(\frac{P}{P^*E}\right) = q \cdot Ex^*\left(\frac{1}{q}\right)$$

  – where I assumed that the prices of internationally traded goods are *set (and sticky)* in producers currency [**producer-currency pricing or PCP**]

How do changes in *q* affect *NX*?

$$NX(q) = Ex(q) - q \cdot Ex^*\left(\frac{1}{q}\right)$$

- Where I omitted the explicit dependence on output for simplicity

- Exports
  - An increase in domestic real ER q (i.e., a real depreciation of domestic currency) increases foreign demand for domestic goods: $\frac{dEx}{dq} > 0$
- Imports
  - How does an increase in q affect *Im*?

How do changes in *q* affect *Im*?

- *Im* denotes the value of imports measured in terms of domestic output:

$$Im = q \cdot Ex^* \left(\frac{1}{q}\right)$$

- Take a derivative

$$\frac{dIm}{dq} = \underbrace{1 \cdot Ex^*}_{\substack{\text{Value effect} \\ >0}} + \underbrace{q\frac{dEx^*}{dq}}_{\substack{\text{Volume effect} \\ <0}}$$

How do changes in *q* affect *NX*?

- Combine $\frac{dEx}{dq}$ *and* $\frac{dIm}{dq}$ *to get* $\frac{dNX}{dq}$

$$\frac{dNX}{dq} = \underbrace{\left(\frac{dEx}{dq} - q\frac{dEx^*}{dq}\right)}_{>0}\underbrace{-Ex^*}_{<0}$$

- Whether NX improves or worsens depends on which effect of a real exchange rate change is dominant
- Two important insights
  - The Marshall-Lerner condition
  - The J-curve

The Marshall-Lerner Condition

The Marshall-Lerner Condition

- Suppose that trade is initially **balanced**

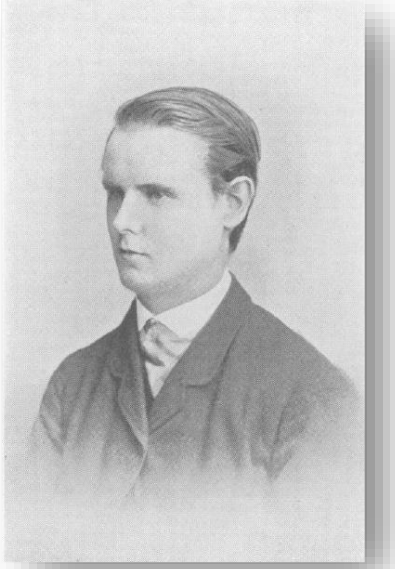
$$Ex = Im = q \cdot Ex^*$$

- The change in NX after a small change in q is

$$\frac{dNX}{dq} = (\eta + \eta^* - 1)Ex^*$$

η and $-\eta^*$ are trade elasticities of Ex and Ex* with respect to the real exchange rate q

- Condition: $\eta + \eta^* > 1$



Alfred Marshall
1842-1924

Abba P. Lerner
1903-1982

Derivation of the M-L Condition

$$\frac{dNX}{dq} = \frac{dEx}{dq} - q\frac{dEx^*}{dq} - Ex^*$$

$$= \underbrace{\frac{dEx}{dq} \cdot \frac{q}{Ex}}_{\eta} \cdot \frac{Ex}{q} - \underbrace{\frac{dEx^*}{dq} \cdot \frac{q}{Ex^*}}_{\eta^*} \cdot q \cdot \frac{Ex^*}{q} - Ex^*$$

$$= \eta\frac{Ex}{q} + \eta^* Ex^* - Ex^*$$

$$= \left(\eta\frac{Ex}{qEx^*} + \eta^* - 1\right)Ex^*$$

Trade Elasticities and Reality

TABLE 17A2-1 Estimated Price Elasticities for International Trade in Manufactured Goods

	η			η^*		
Country	Impact	Short-run (6 months)	Long-run (12 months)	Impact	Short-run	Long-run
Austria	0.39	0.71	1.37	0.03	0.36	0.80
Belgium	0.18	0.59	1.55	—	—	0.70
Britain	—	—	0.31	0.60	0.75	0.75
Canada	0.08	0.40	0.71	0.72	0.72	0.72
Denmark	0.82	1.13	1.13	0.55	0.93	1.14
France	0.20	0.48	1.25	—	0.49	0.60
Germany	—	—	1.41	0.57	0.77	0.77
Italy	—	0.56	0.64	0.94	0.94	0.94
Japan	0.59	1.01	1.61	0.16	0.72	0.97
Netherlands	0.24	0.49	0.89	0.71	1.22	1.22
Norway	0.40	0.74	1.49	—	0.01	0.71
Sweden	0.27	0.73	1.59	—	—	0.94
Switzerland	0.28	0.42	0.73	0.25	0.25	0.25
United States	0.18	0.48	1.67	—	1.06	1.06

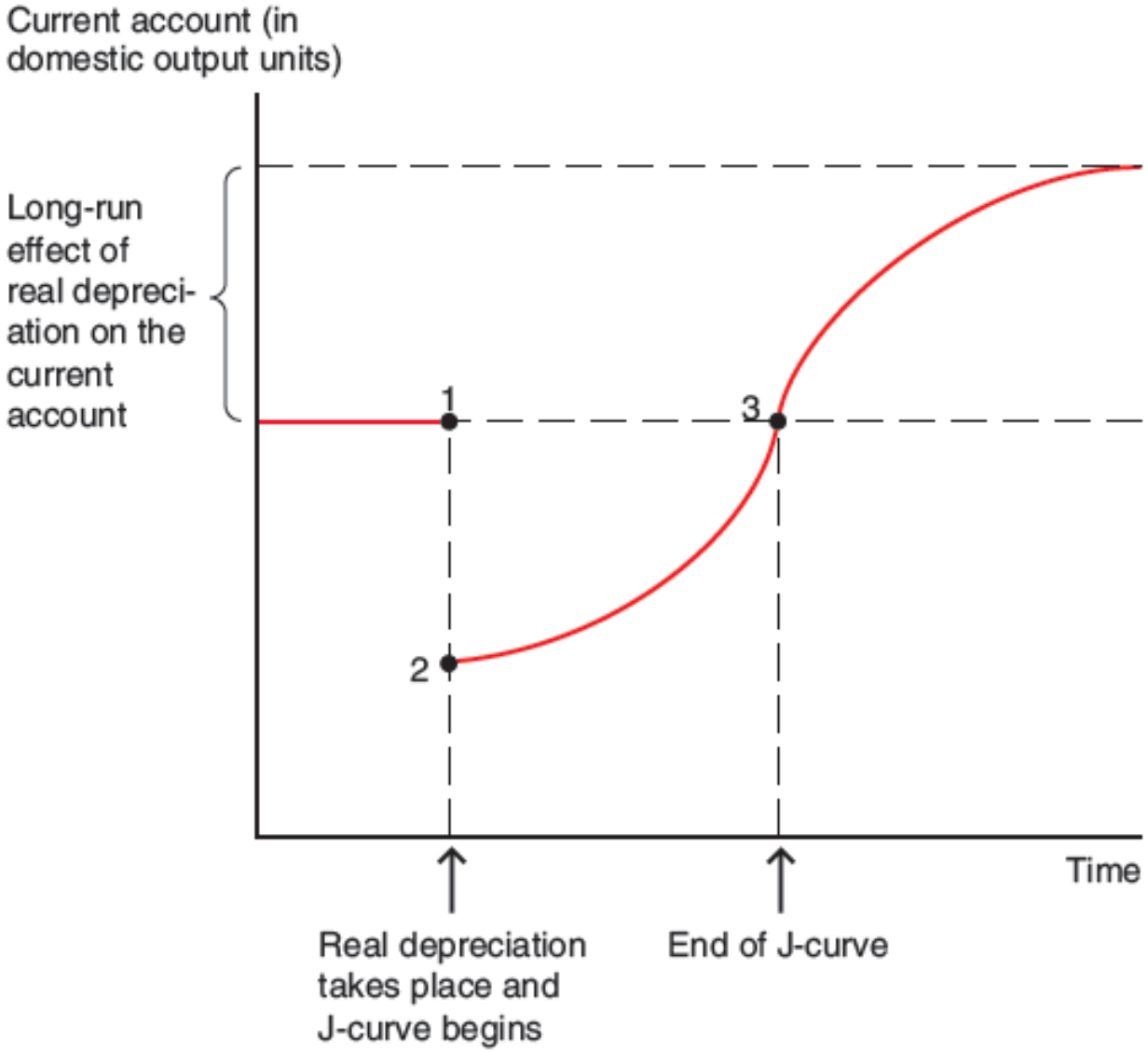
Source: Estimates are taken from Jacques R. Artus and Malcolm D. Knight, *Issues in the Assessment of the Exchange Rates of Industrial Countries*. Occasional Paper 29. Washington, D.C.: International Monetary Fund, July 1984, table 4. Unavailable estimates are indicated by dashes.

The M-L condition does not hold on "impact," but holds in the "short and long runs."

The J-Curve

How do changes in *q* affect *NX*?

The J-Curve



- The curve describes the time lag with which a real currency depreciation improves the NX
- **Most of the time in our class, we will assume that NX responds positively to q**

How do changes in E affect *NX*?

- Recall that under the **PCP**

$$NX(q) = Ex(q) - qEx^*\left(\frac{1}{q}\right)$$

- Real exchange rate

$$q = \frac{P^*E}{P}$$

- When prices P^* and P are sticky, then

$$\frac{\Delta q}{q} = \frac{\Delta E}{E}$$

implying the E affects NX in the same way q does.

How do changes in Y^d affect *NX*?

- An increase in disposable income (Y^d) worsens *NX*
  - Intuition: people at home start consuming more foreign goods (imports go up)

- Similarly, an increase in foreign disposable income ($Y^{d,*}$) improves the trade balance *NX*

Aggregate Demand

The AD equation

- The four components of aggregate demand are combined to get the total aggregate demand

$$D = D(q, Y^d, I, G) = C(Y^d) + I + G + NX\left(q, Y^d\right)$$

What affects Aggregate Demand (AD)?

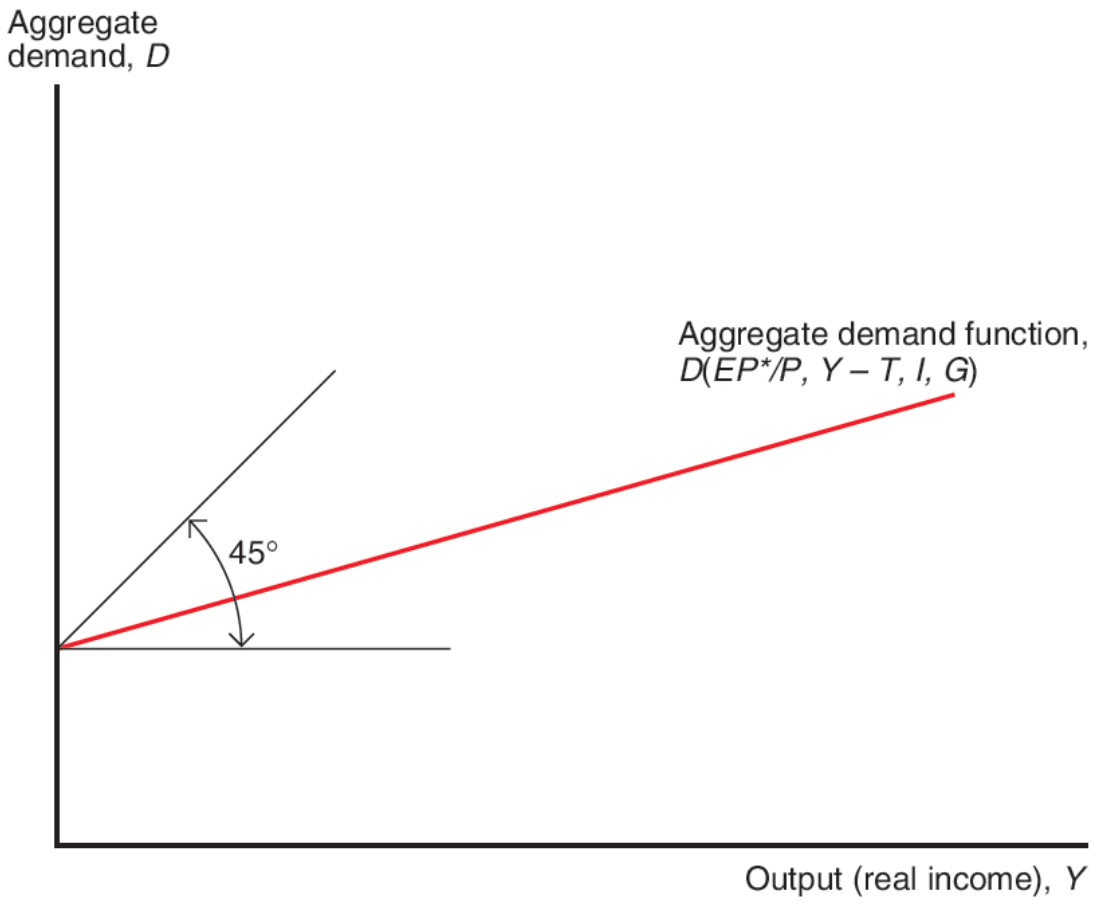
$$D = C(Y^d) + I + G + NX\left(q, Y^d\right)$$

1. $\frac{dD}{dq} > 0$ (real depreciation of home currency increases AD)

2. $\frac{dD}{dY^d} > 0$ (an increase in dispos. income increases AD) Why?

3. $\frac{dD}{dI} > 0, \frac{dD}{dG} > 0$

AD and Disposable Income



- $0 < \frac{dD}{dY^d} < 1$ because consumption reacts less than one-to-one
- $D\left(Y^d = 0\right) > 0$ because of government, investment, and foreign demand

Goods Market Equilibrium

Goods Market Equilibrium in the Short Run

- When output (and labor) market is in equilibrium, output equals aggregate demand

$$Y = D\left(\frac{EP^*}{P}, Y - T, I, G\right)$$

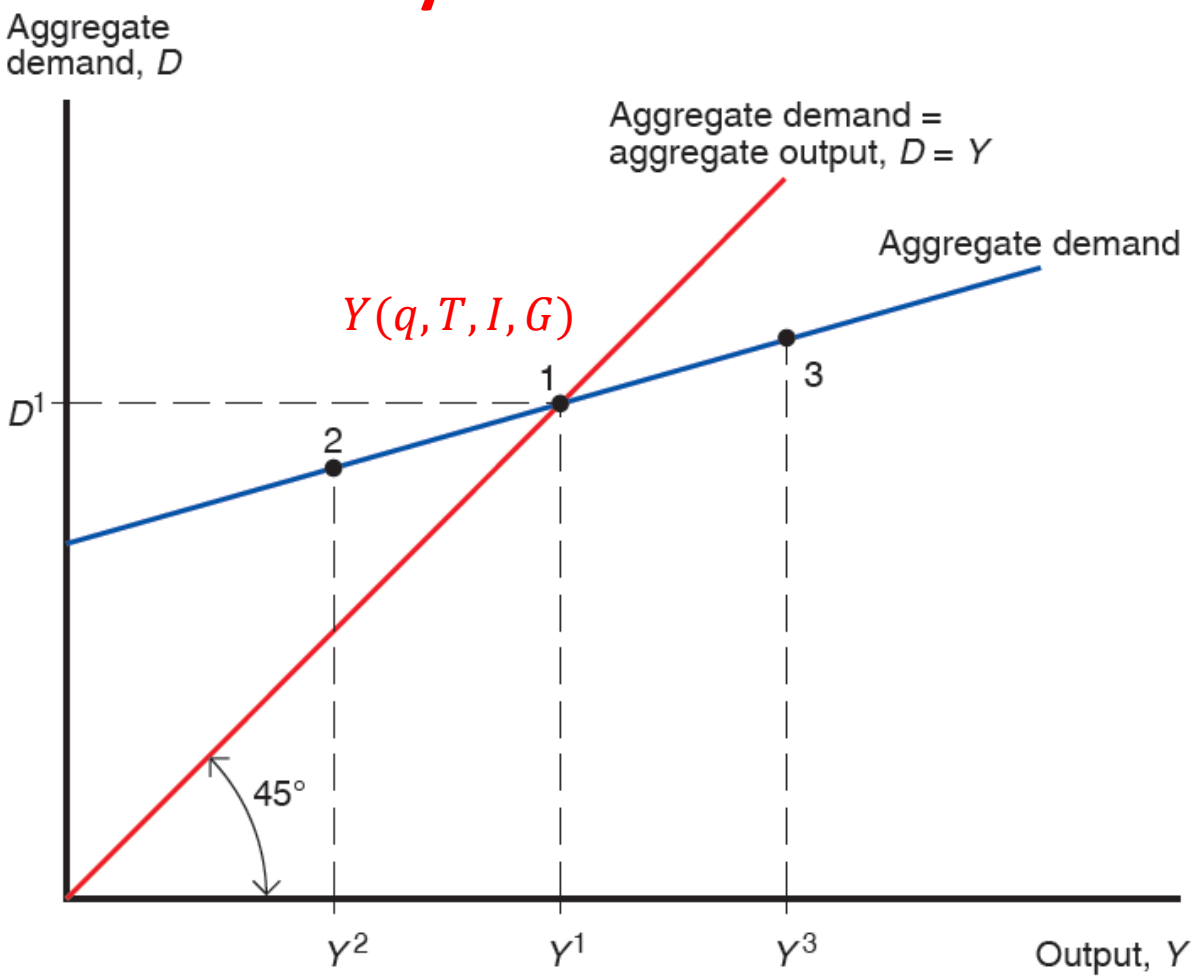
  – Labor market equilibrium: output equals households income

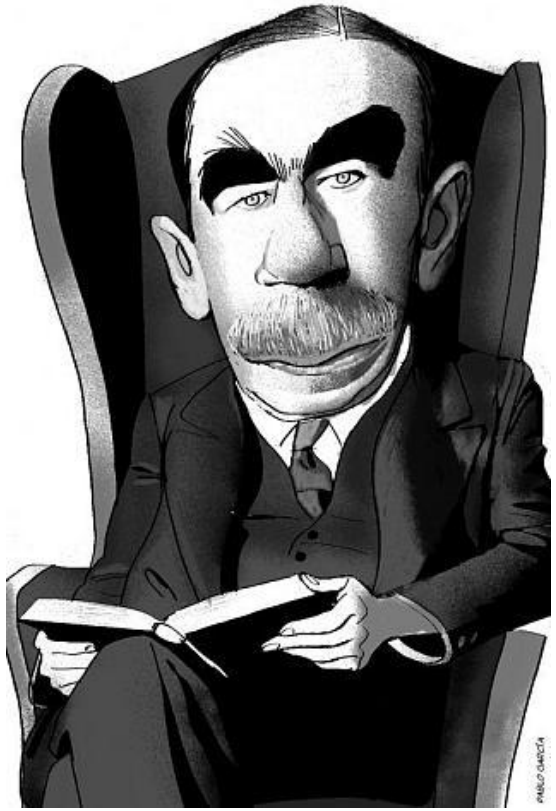
  – Note that we ignored here the difference between GNDI and GDP by assuming (for simplicity) that NIFA=NUT=0

NUT = Net Unilateral Transfers

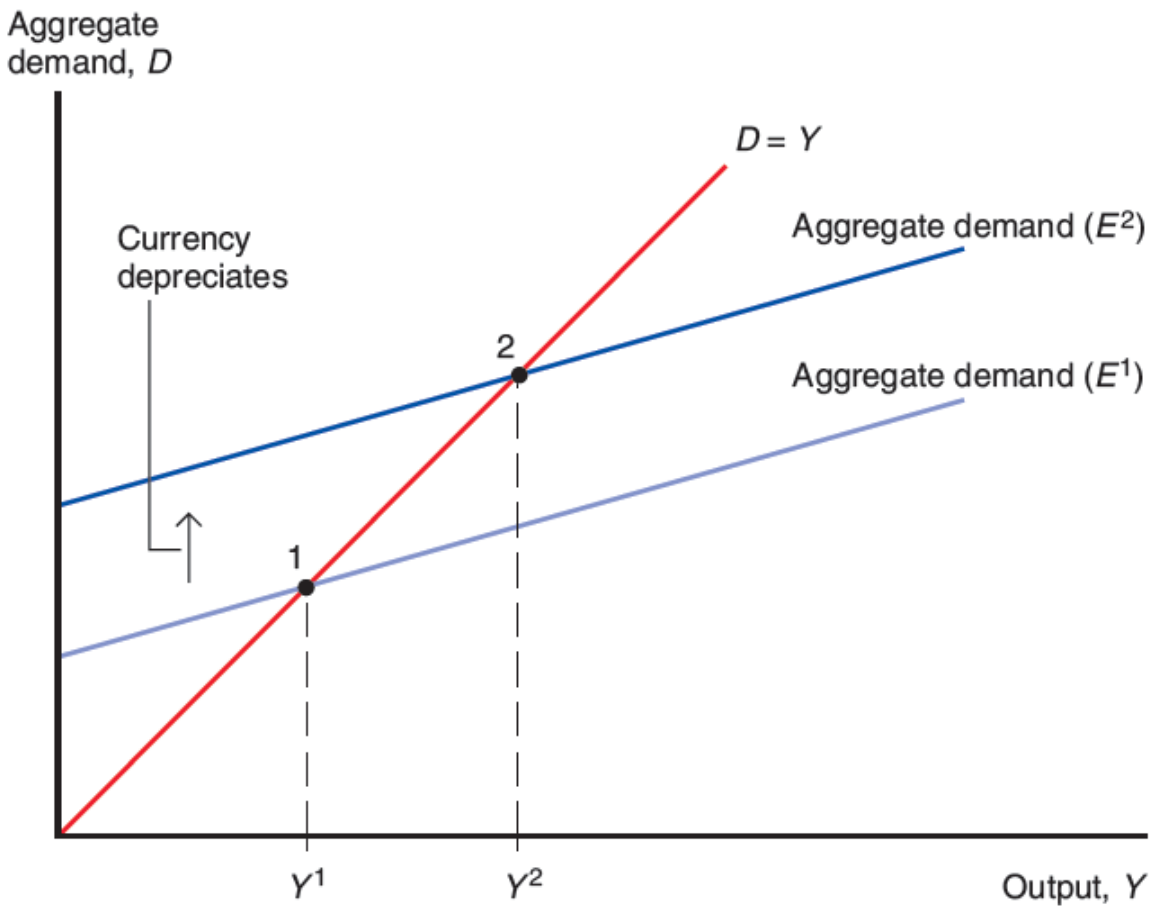
Goods Market Equilibrium in the Short Run

Keynesian cross

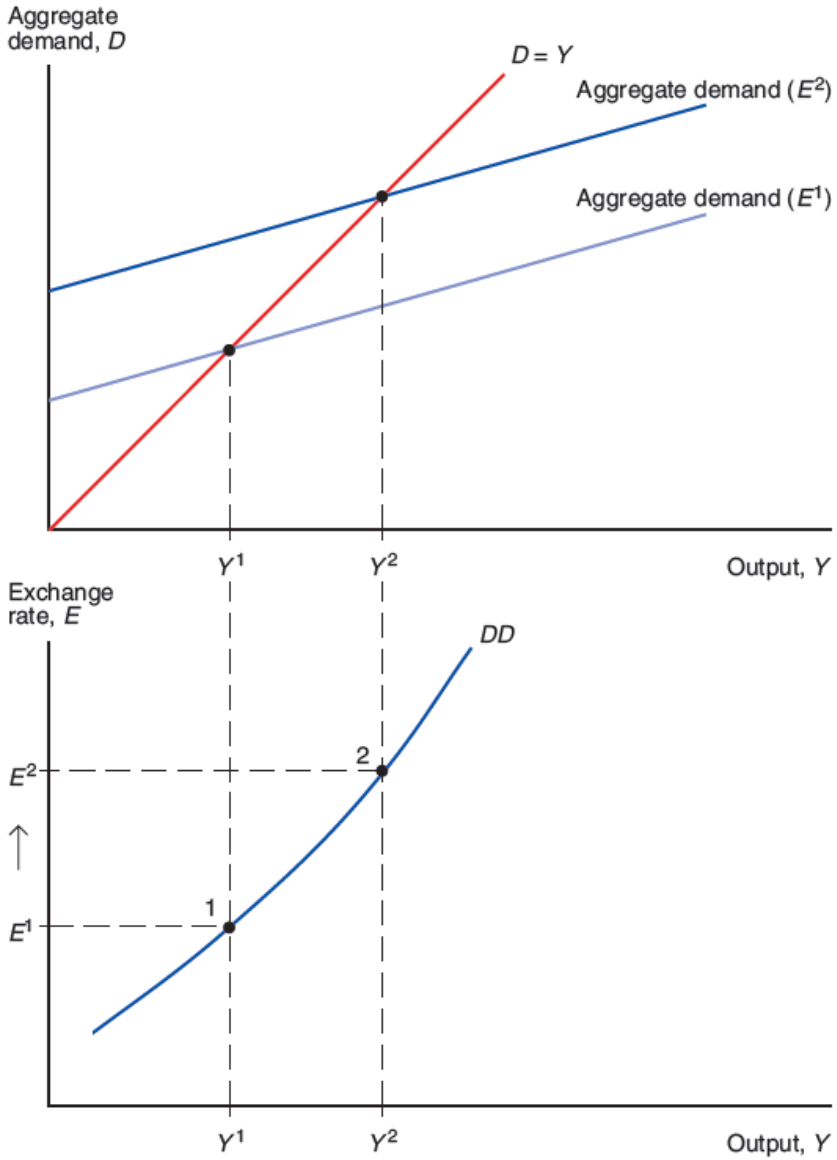




How does domestic currency depreciation affect goods market equilibrium?



The DD Schedule



$$Y = D\left(\frac{EP^*}{P}, Y - T, I, G\right)$$

$$Y = Y\left(\frac{EP^*}{P}, T, I, G\right)$$

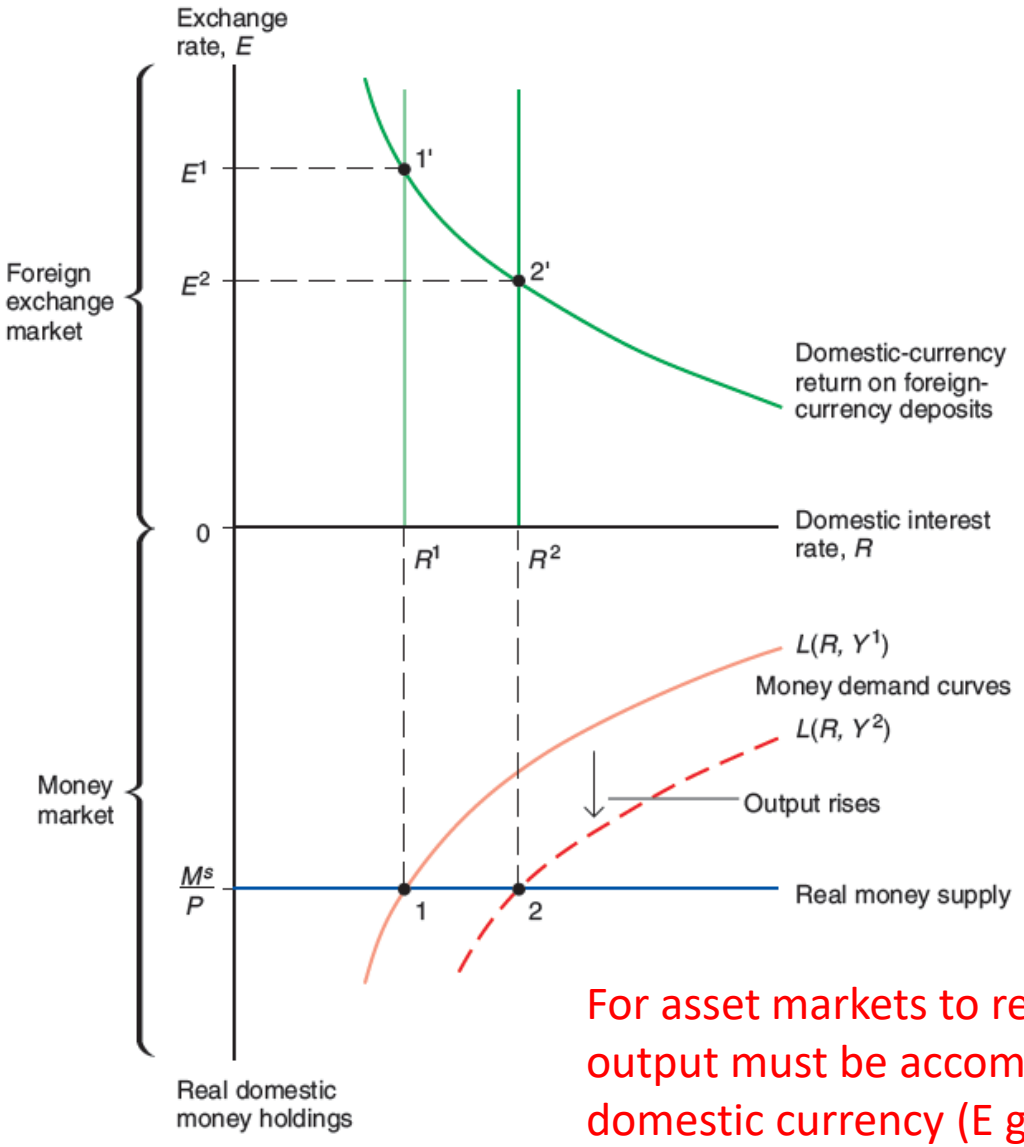
The AA Schedule

The AA Schedule

- The DD schedule only links Y to E
  - We need one more relation to solve for Y and E

- The AA Schedule
  - shows all combinations of exchange rate and output that are consistent with equilibrium in the domestic money market and the foreign exchange market.

The AA Schedule

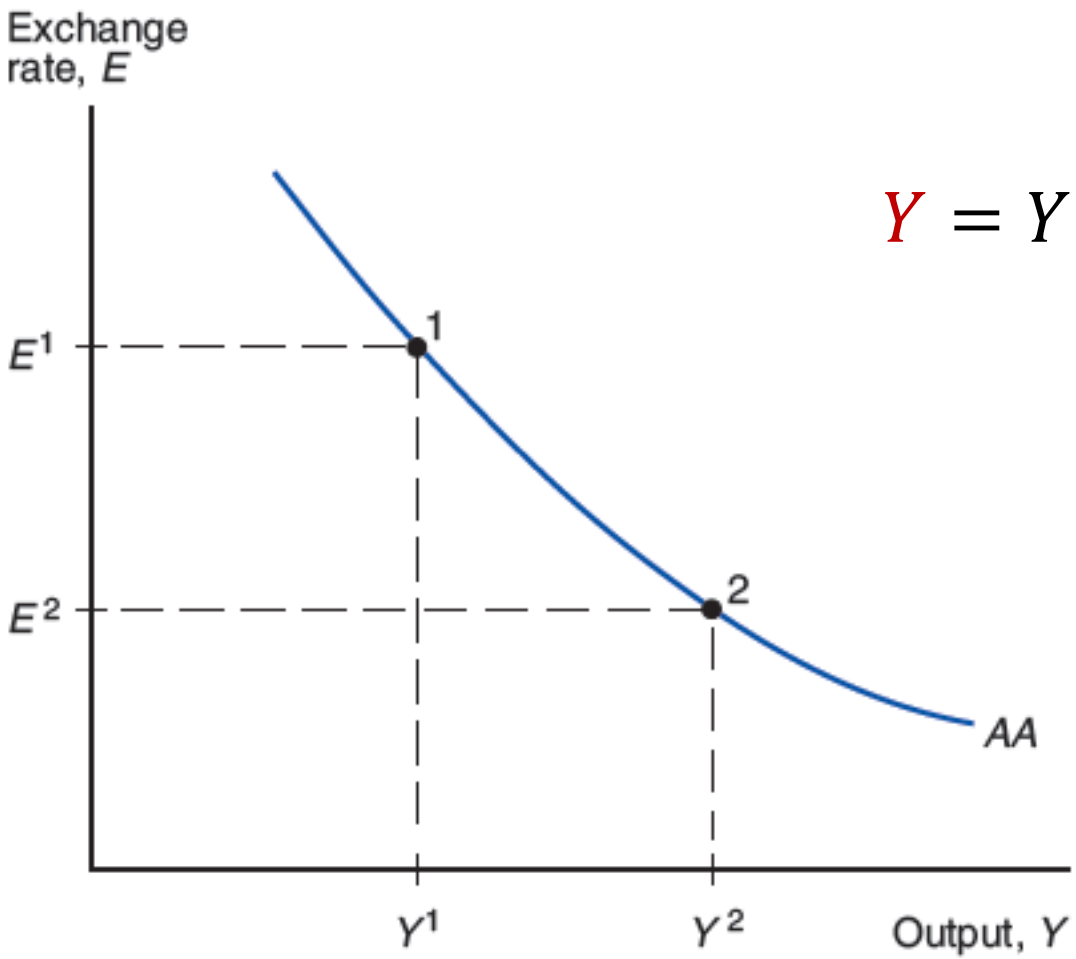


$$R = \bar{R}^* + \frac{\bar{E}^e - E}{E}$$

$$\frac{\bar{M}^s}{\bar{P}} = L(R, Y)$$

For asset markets to remain in equilibrium a rise in domestic output must be accompanied by an appreciation of the domestic currency (E goes down)

The AA Schedule

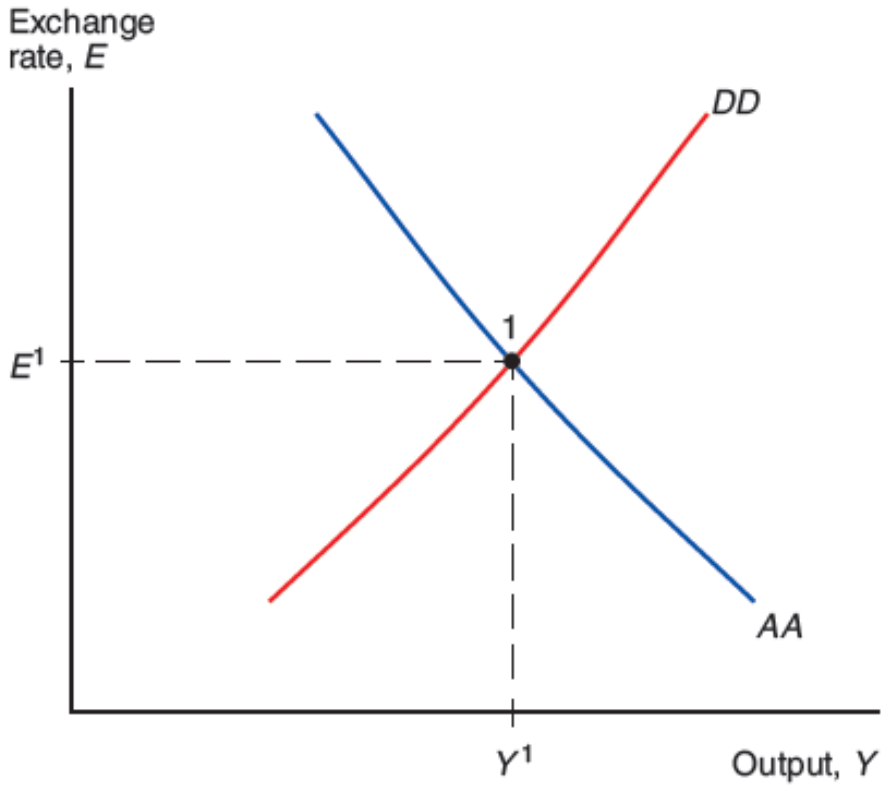


$$Y = Y(E, E^e, R^*, P, M^s)$$

Short-Run Equilibrium

Short-Run Equilibrium

- A short-run equilibrium for the economy as a whole must bring equilibrium simultaneously in the output and asset markets



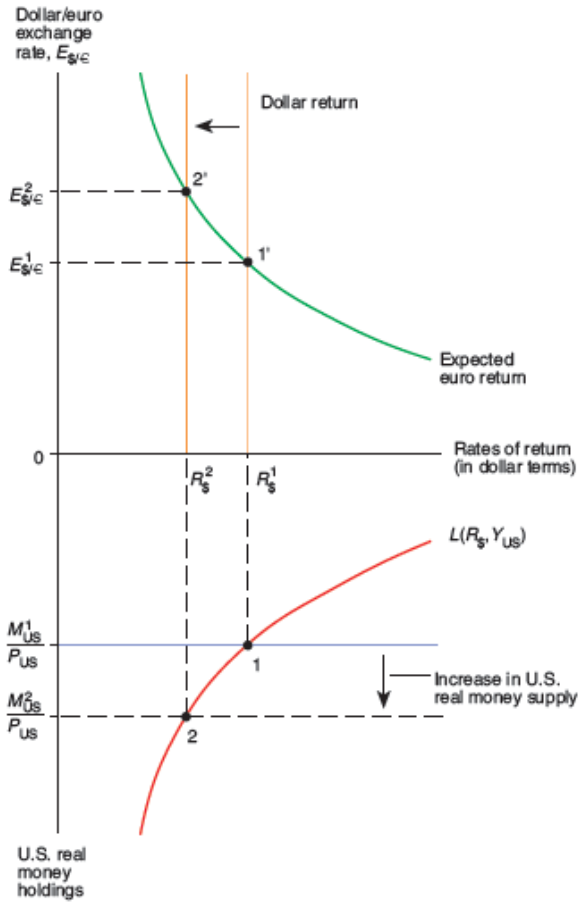
Policy Experiments

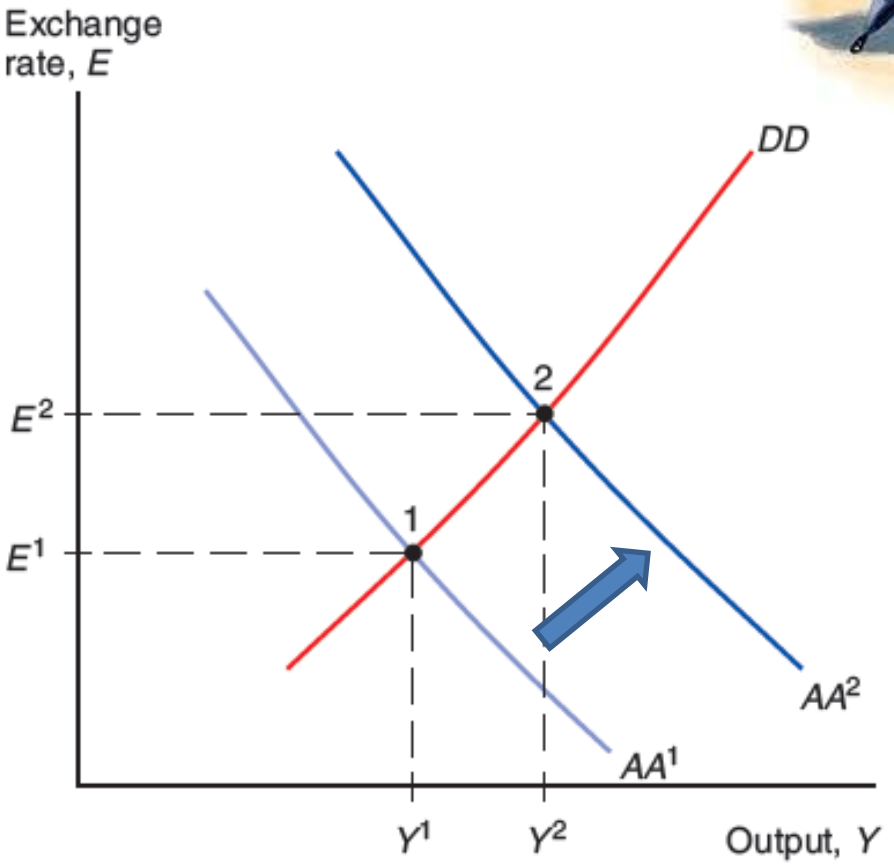
Temporary Changes in Monetary and Fiscal Policy

- Monetary policy
  - changes in the money supply
- Fiscal policy
  - changes in government spending or taxes
- Assumptions: temporary policy changes don't affect
  1. expected exchange rate: E^e
  2. foreign variables: R^* and P^*
  3. domestic price level: P

Exercise 1

Temporary Increase in Money Supply





Exercise 1

A Temporary Increase in Money Supply

Note that

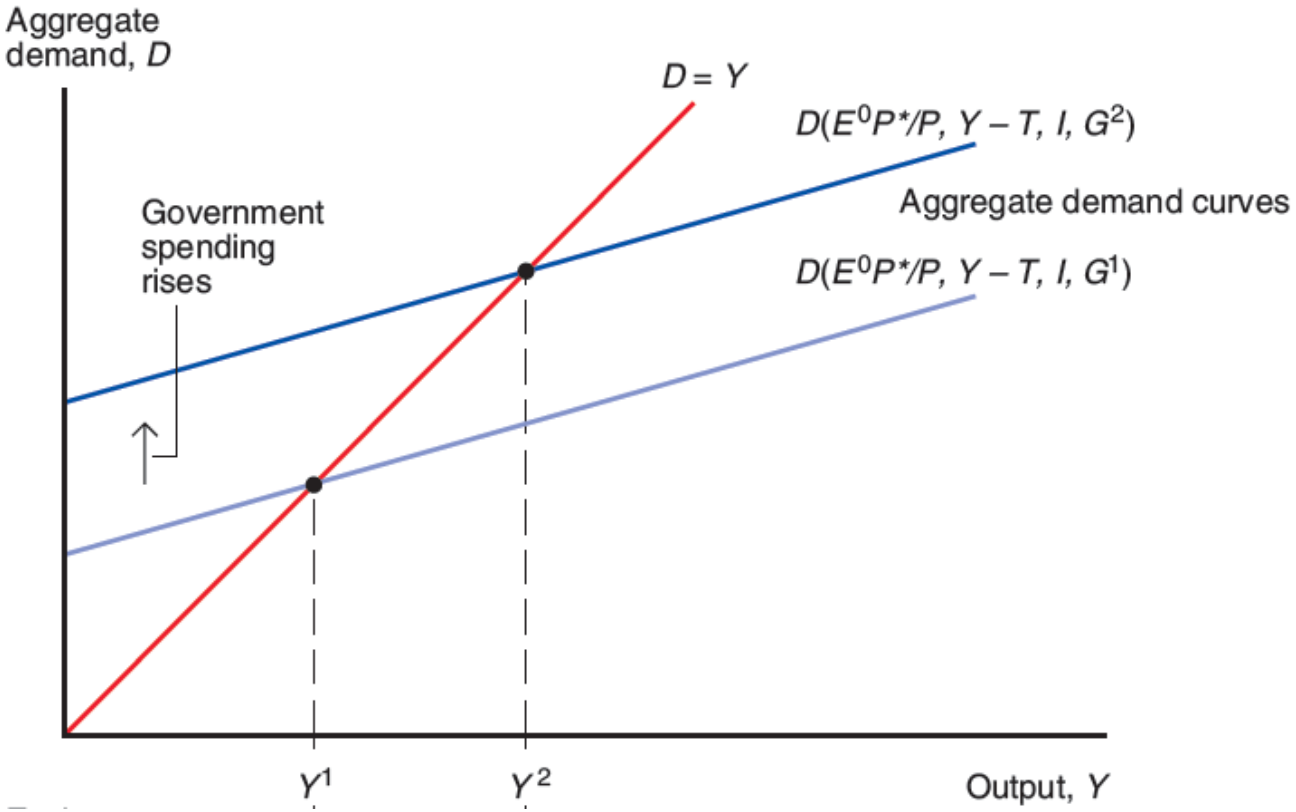
- An increase in the money supply raises the economy's output!

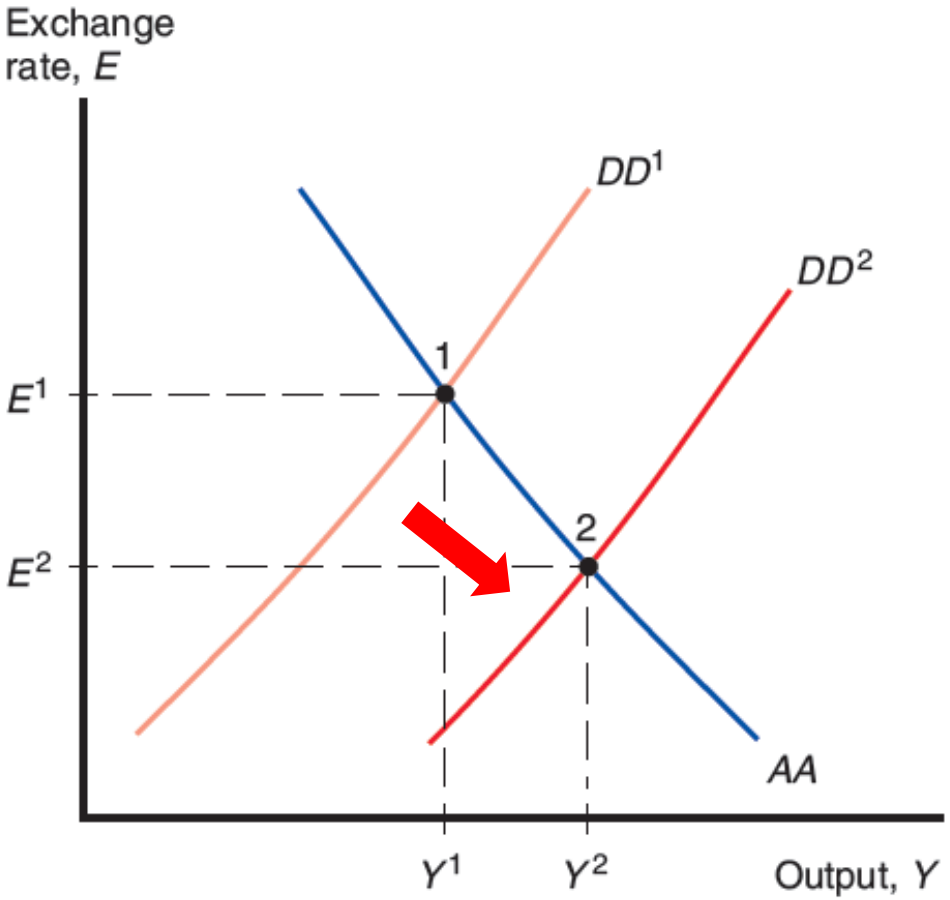
- But the channel of transmission is distinct from the standard closed-economy one, **which we assumed away for simplicity***
  - expansionary monetary policy lowers the nominal interest rate, which leads to an immediate depreciation of the currency

*Note that by the standard closed-economy mechanism I mean an increase in consumption and investment demand following a decline in the interest rate

Exercise 2

A Temporary Change in Fiscal Policy





Exercise 2

Temporary Changes in Fiscal Policy

- An increase in government spending, a cut in taxes, or some combination of the two (i.e., expansionary fiscal policy) raises output

- The expansionary effect is partially offset by an appreciation of the currency

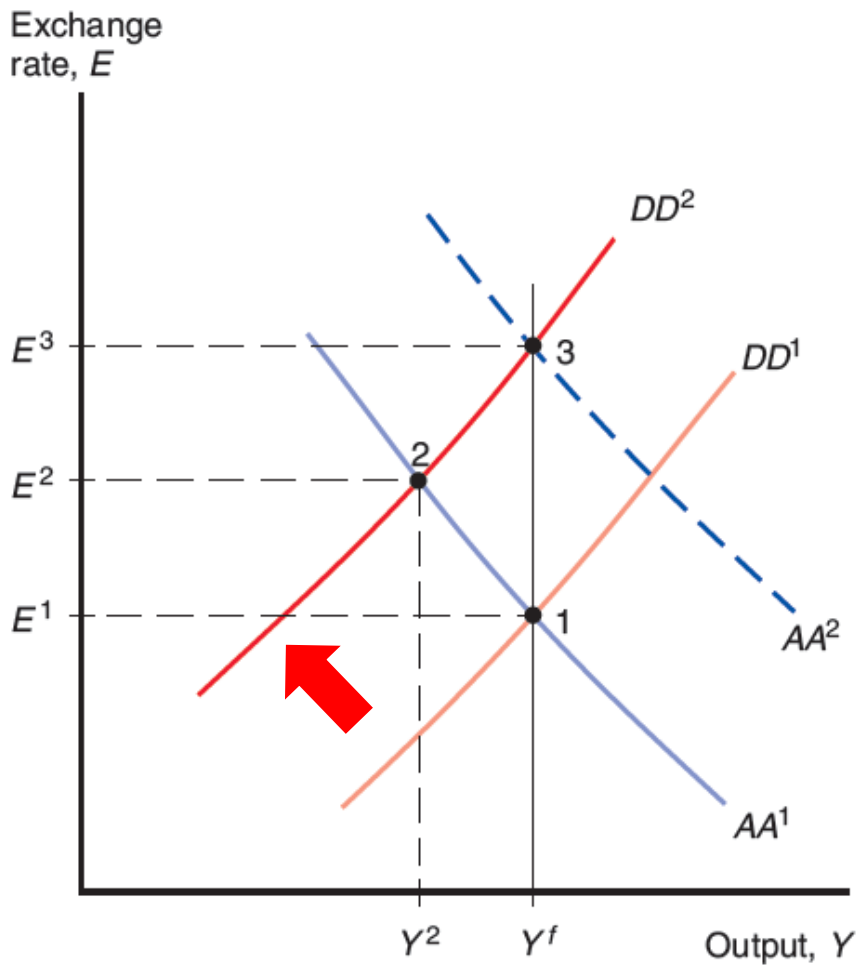
Business Cycles Stabilization

- Recessions
  - Temporary disturbances that lead to recessions can be offset through monetary or fiscal policies

- Booms
  - Temporary disturbances that lead to overemployment can be offset through contractionary monetary or fiscal policies

Example 3

A Temporary Fall in a World Demand for Domestic Products



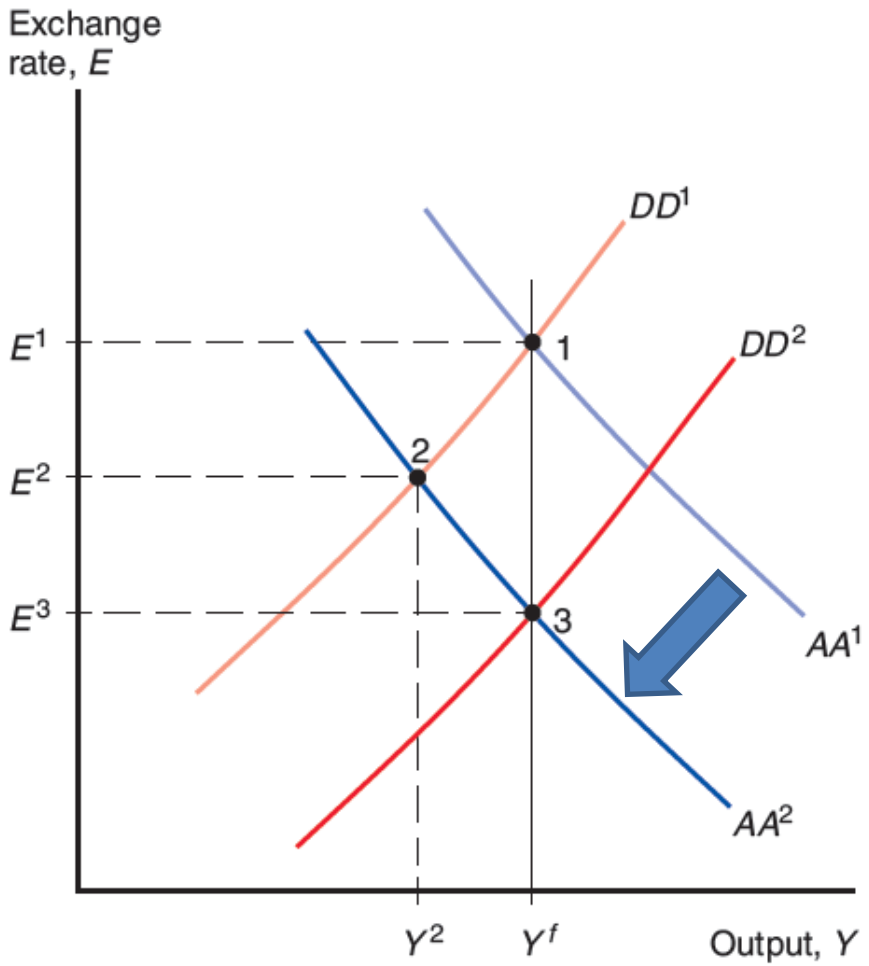
The shock moves the economy from point 1 to point 2.

A recession can be avoided by

1. a fiscal expansion (return to point 1)
2. a monetary expansion (point 3)

Example 4

A Temporary Money Demand Increase



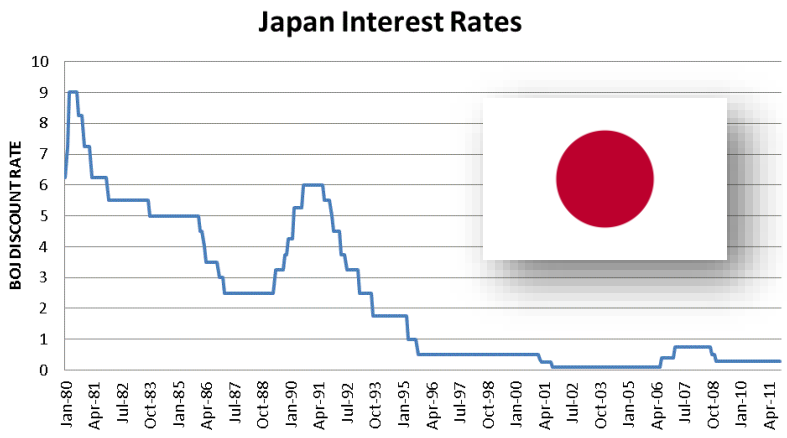
The shock moves the economy from point 1 to point 2.

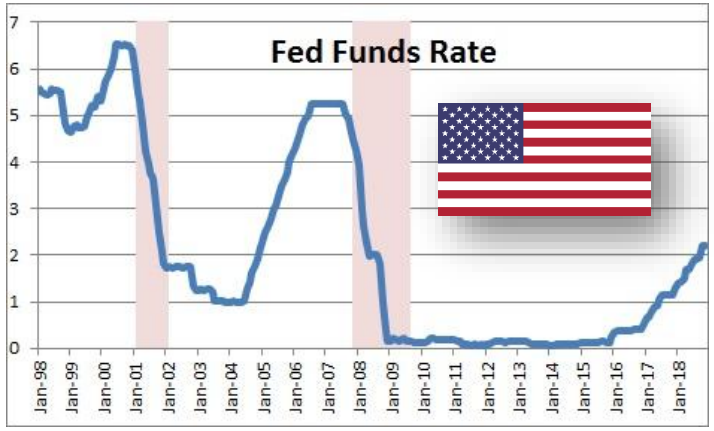
A recession can be averted by

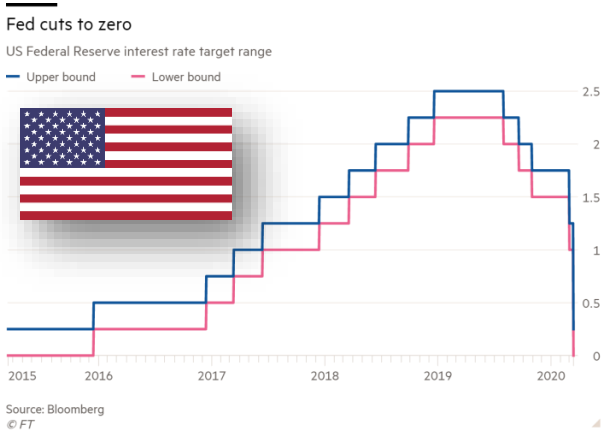
1. A fiscal expansion (point 3)
2. A monetary expansion (point 1)

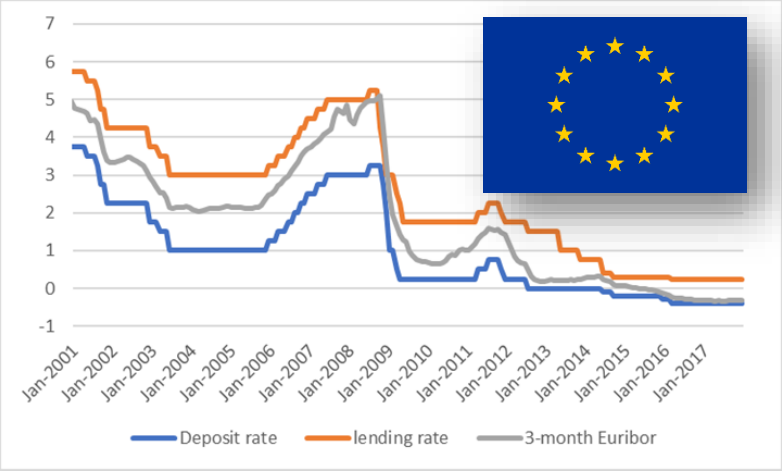
Liquidity Trap

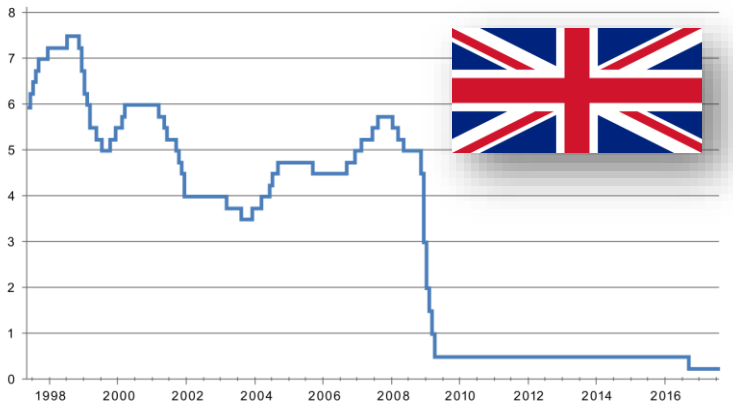
Liquidity Trap











Liquidity Trap

- When the nominal interest rate reaches 0 central bank cannot lower interest rates any further. Why?
  - In fact, central banks can reduce the interest rate slightly below zero but not too much

- Conventional monetary policy becomes ineffective

Liquidity Trap

- The UIP

$$R = 0 = R^* + \frac{E^e - E}{E}$$

- Exchange rate does not depend on monetary policy

$$E = \frac{E^e}{1 - R^*}$$